Reflective Journal

1. Understanding the ML Workflow

Working through the machine learning workflow revealed how each step logically builds toward evaluating model performance rather than just being a sequence of arbitrary tasks. The most critical insight was understanding why we split data into training (80%) and testing (20%) sets: the training set allows the model to learn underlying patterns, while the test set acts as an unbiased evaluation on unseen data to ensure the model isn't simply memorizing inputs. Additionally, preparing data such as encoding the categorical 'Sex' column into binary numerical values (0 and 1) highlighted that machine learning algorithms are fundamentally mathematical engines that require clean, numeric input to function properly.

2. Regression vs. Classification

The lab clearly demonstrated the fundamental distinction between regression and classification problems based on the nature of their target variables. Regression tasks, such as predicting a passenger's continuous ticket 'Fare', require metrics like Mean Squared Error (MSE) to quantify how far off predictions are from actual values on a continuous scale. In contrast, classification tasks, like predicting categorical 'Survived' outcomes (0 or 1), rely on Accuracy Score to measure the proportion of correct discrete guesses. Applying accuracy to a regression problem like fare prediction would be ineffective because hitting an exact dollar amount down to the penny is

extremely rare, making continuous error measurement essential for evaluating numeric predictions.

3. Demystifying Model Weights (.coef_)

Inspecting the .coef_ attribute of the trained Linear Regression model demystified how the algorithm transforms input features into final predictions. These coefficients represent the mathematical weights assigned to each feature, indicating both the strength and direction of their influence on the target variable. For instance, a negative coefficient assigned to passenger class (Pclass) aligns with domain intuition, as moving to a higher numerical class (e.g. 3rd class) corresponds to lower ticket fares. Understanding these weights converts the model from a "black box" into an interpretable system where individual feature contributions can be measured and validated.

4. Key Takeaways & Next Steps

This exercise reinforced that model metrics must be translated into real-world units to be meaningful, as seen when taking the square root of the MSE (approximately 58.01) to realize predictions were off by about 58 on average. Moving forward, improving model performance will likely require incorporating richer feature sets beyond just age and class, alongside learning how to handle missing data without discarding valuable rows. Ultimately, this lab shifted my perspective from viewing machine learning as writing code to viewing it as a structured process of preparing data, selecting appropriate algorithms, and critically interpreting model outputs.